Air Fryer: Demand Estimation & Market Analysis

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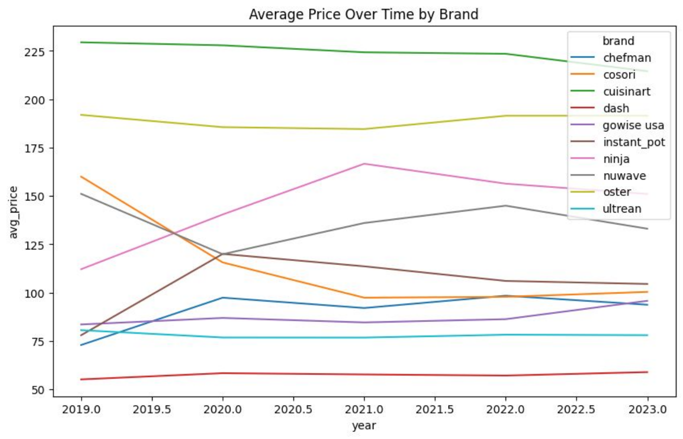
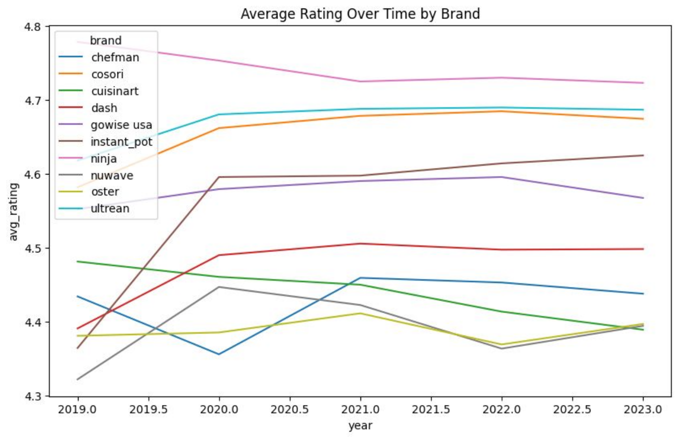
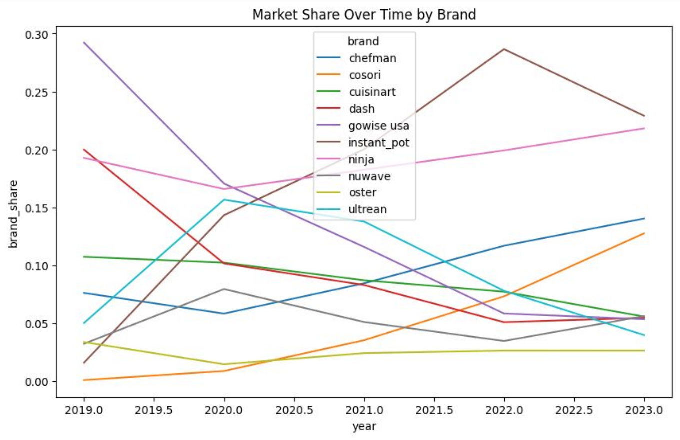
5/5/2026

Through our analysis, we have determined that the Air Fryer market is differentiated through pricing and product features rather than large differences in perceived quality. We notice this market being highly rated across all brands, with the higher-rated brands and compact-features having the largest percentage of the market shares.

Now, we will look at our findings and how we came to these conclusions.







brand	compact_share	dual_basket_share	oven_style_share	rotisserie_share	window_share
chefman	0.961595	0.013462	0.596962	0.370490	0.363081
cosori	0.996784	0.000000	0.029952	0.024076	0.000000
cuisinart	0.995902	0.000000	0.913059	0.000000	0.000000
dash	0.999481	0.000000	0.890098	0.000000	0.000000
gowise usa	0.999877	0.000000	0.183790	0.183569	0.001320
instant_pot	0.859655	0.000030	0.674784	0.102362	0.003012
ninja	0.992084	0.002488	0.100477	0.000000	0.000000
nuwave	0.995431	0.006622	0.542646	0.027123	0.000000
oster	1.000000	0.000000	0.864518	0.000000	0.000000
ultrean	1.000000	0.000000	0.830181	0.000000	0.000000

Demand Model & Feature Information

	feature	coefficient
0	avg_price	-0.037668
1	avg_rating	0.287517
2	compact_share	9.815304
3	dual_basket_share	-9.509686
4	oven_style_share	1.941774
5	rotisserie_share	-5.674054
6	window_share	12.880298
7	brand_cosori	2.551946
8	brand_cuisinart	6.422436
9	brand_dash	0.176655
10	brand_gowise usa	3.938996
11	brand_instant_pot	4.626260
12	brand_ninja	5.838705
13	brand_nuwave	3.544883
14	brand_oster	3.928074
15	brand_ultrean	0.942399
16	year_2020	0.119071
17	year_2021	0.041900
18	year_2022	-0.098860
19	year_2023	-0.003307

We used the coefficients of the features to determine the demand in the market for each.

Looking at the estimated average price coefficient at the top of the table, we can see that it is negative (-0.037668). This tells us that our model is reliable, as it matches the intuition that as the price increases, the demand for it decreases.

Applying this same logic, we can look at the features of the air fryers to see which are of highest demand and take up the largest portion of the market share. Due to the positive coefficients of compact, oven style, and window, we can conclude that those are in higher demand than dual basket and rotisserie. For product demand and overall success, this implies that focusing on these features, especially window and compact, sets the brand up for more success.

Next, we looked at the demands for different brands. For dummy coefficients to work, we must drop one brand to use as a baseline, which was Chefman. Relative to Chefman, all brands had more market share, and therefore more demand. Those with the highest coefficients, like Cuisinart (6.422) and Ninja (5.839) show the highest demand.

Finally, we look at the year. Again, we dropped a year as a baseline, which was 2019. Relative to 2019, years 2020 and 2021 had slightly higher market share, so slightly higher demand. Since, the demand has gone down. This shows that the demand was highest during the pandemic era, then softened towards 2022.

Estimated Average Unit Costs

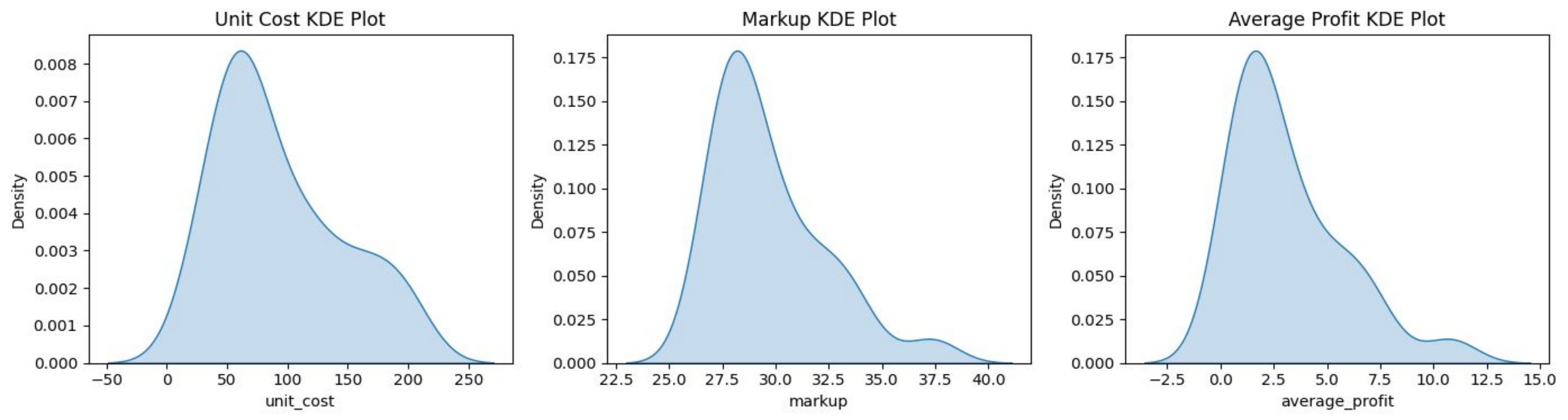
brand	unit_cost
cuisinart	194.896119
oster	161.804969
ninja	112.488470
nuwave	109.053549
cosori	86.283321
instant_pot	71.903119
chefman	61.569466
gowise usa	56.301040
ultrean	48.785765
dash	27.936578

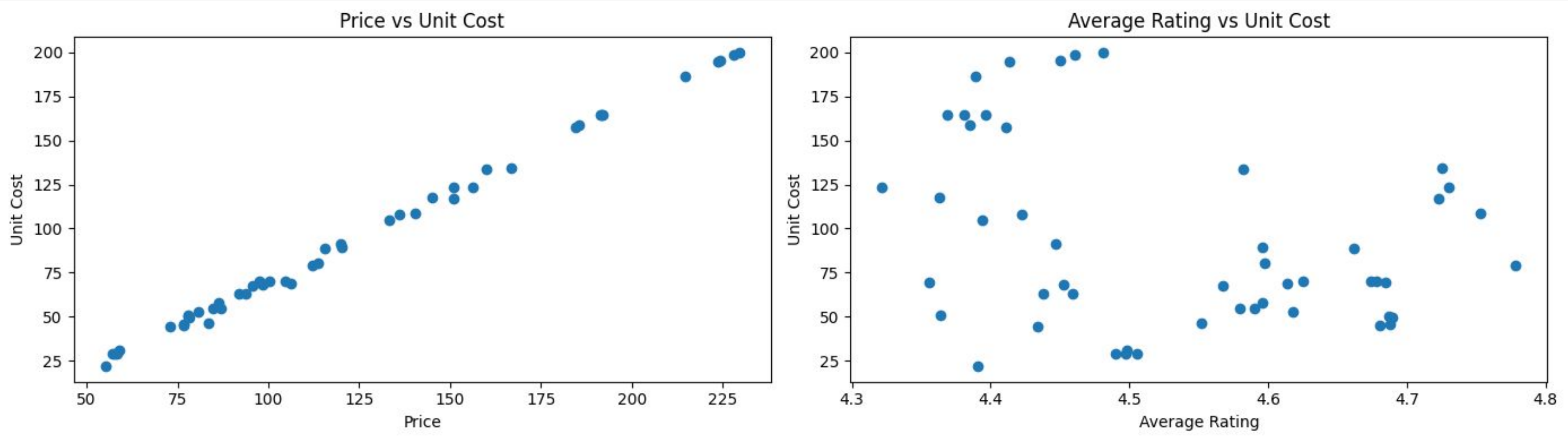
Estimated Average Markups

brand	markup
ninja	32.854071
instant_pot	32.558234
gowise usa	31.153741
dash	29.542349
chefman	29.368946
ultrean	29.328841
cuisinart	29.050974
cosori	27.984628
nuwave	27.970835
oster	27.226315

Estimated Average Share-Weighted Profits

brand	average_profit
ninja	6.306093
instant_pot	6.010256
gowise usa	4.605764
dash	2.994372
chefman	2.820968
ultrean	2.780864
cuisinart	2.502997
cosori	1.436651
nuwave	1.422858
oster	0.678338


Unit Cost KDE Plot
Density
0.008
0.007
0.006
0.005
0.004
0.003
0.002
0.001
0.000
−50
0
50
100
150
200
250
unit_cost
Markup KDE Plot
Density
0.175
0.150
0.125
0.100
0.075
0.050
0.025
0.000
22.5
25.0
27.5
30.0
32.5
35.0
37.5
40.0
markup
Average Profit KDE Plot
Density
0.175
0.150
0.125
0.100
0.075
0.050
0.025
0.000
−2.5
0.0
2.5
5.0
7.5
10.0
12.5
15.0
average_profit


Price vs Unit Cost
Unit Cost
200
175
150
125
100
75
50
25
50
75
100
125
150
175
200
225
Price
Average Rating vs Unit Cost
Unit Cost
200
175
150
125
100
75
50
25
4.3
4.4
4.5
4.6
4.7
4.8
Average Rating

Based on our overall analysis of these brands, product features, and market estimations, we believe that the two strongest brands are Ninja and Instant Pot. They not only have consistently high ratings, but also market share, estimated markups and estimated profits. The ratings show us that consumers like the product, which will likely get the brands more support. In addition, the larger market shares show that they are in higher demand. Lastly, the combination of higher markups and larger market share show us that the brands are dominating the market while getting the most value.

Next, we would want to investigate what would happen if these weaker brands with similar features and fair ratings were to lower their prices. Would competition with these stronger brands increase, or remain the same?

Thank you!